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AGILE PROJECT BLUEPRINT

Student Record & Academic Management Platform

UNIT 2 MINI-PROJECT | SCRUM FRAMEWORK | RELEASE 1.0

1. Product Vision & Scope

Vision Statement

"Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration, and reporting through a scalable web-based application."

Primary Objectives: Centralize academic data, automate attendance tracking, streamline result computation, and provide role-based self-service capabilities for students, faculty, and administrators.

2. Scrum Team Structure & Roles

Product Owner

- Defines product vision & features.
- Prioritizes the Product Backlog.
- Represents stakeholder interests.
- Accepts/rejects completed work.

Scrum Master

- Facilitates Scrum ceremonies.
- Removes project impediments.
- Ensures Agile practices are followed.
- Supports team collaboration.

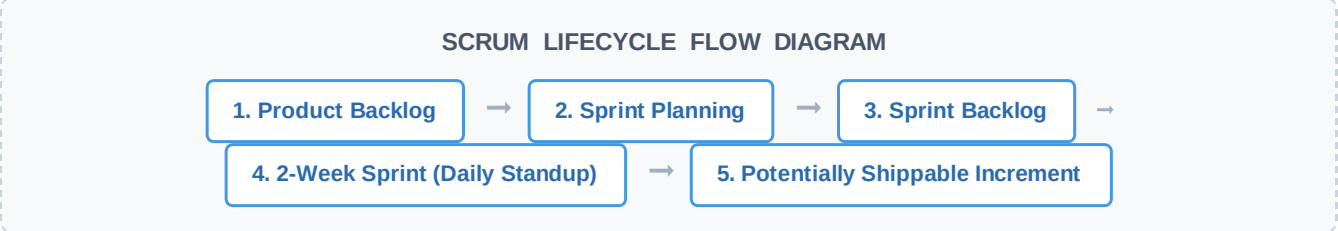
Development Team

- Architects & designs application.
- Writes clean, modular code.
- Performs unit & system testing.
- Demonstrates completed features.

3. Functional Epics Breakdown

Epic ID & Module	Core Features & Deliverables
Epic 1: Student Management	Student Registration, Profile Management, Search, Update, Deletion
Epic 2: Course Management	Add Course, Update Course Details, Assign Students, View Course Roster
Epic 3: Attendance	Record Lecture Attendance, View Student Attendance, Generate Alerts
Epic 4: Marks & Performance	Enter Marks, Update Scores, Calculate GPA/Grades, Generate Result Sheets
Epic 5: Security & Auth	User Login/Logout, Password Reset, Role-Based Access Control (RBAC)
Epic 6: Notifications	Attendance Alerts, Examination Notifications, Grade Release Alerts
Epic 7: Reporting	Student Summary Reports, Attendance Reports, Performance Analytics

4. Agile Development Lifecycle Workflow



5. Key User Stories & Acceptance Criteria

US-01: User Authentication

"As a Registered User, I want to log in using my credentials so that I can access my personalized dashboard securely."

- Login page loads successfully with SSL encryption.
- Registered users can log in with valid credentials.
- Invalid login credentials trigger an clear error notification.
- Successful authentication redirects user based on role (Admin, Faculty, Student).
- Session times out after 15 minutes of inactivity.

US-02: Record Daily Attendance

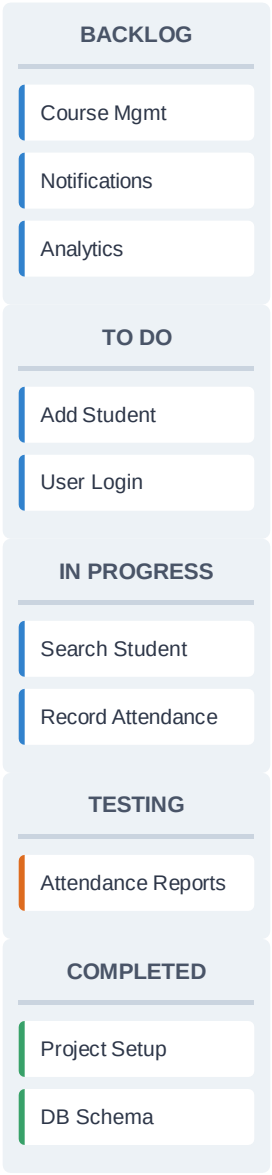
"As a Faculty Member, I want to record student attendance so that attendance records remain accurate and updated."

- Faculty can select course code and active lecture date.
- System presents student list with quick toggle buttons (Present/Absent/Excused).
- Attendance submission updates system records immediately.
- Automated attendance summary report is generated for administrative review.

6. Prioritized Product Backlog & Estimation

Priority	Epic Module	User Story / Feature	Complexity Scale	Story Points
High	Authentication	User Login & Session Management	Moderate	**3**
High	Student Management	Add New Student Record	Moderate	**3**
High	Student Management	Search & Filter Students	Easy	**2**
High	Attendance	Record Lecture Attendance	Difficult	**5**
Medium	Course Management	Add & Configure Course	Moderate	**3**
Medium	Reporting	Generate Attendance Report	Difficult	**5**
Medium	Notifications	Automated Attendance Alerts	Moderate	**3**
Low	Dashboard	Academic Performance Analytics	Very Difficult	**8**

7. Sprint Tracking & Scrum Board Diagram



8. Agile Framework Comparative Evaluation

Framework	Core Characteristics	Suitability for Project
Scrum	Time-boxed Sprints, defined roles (PO/SM/Dev), clear ceremonies.	**Optimal** — Fits project scope & multi-feature releases.
Kanban	Continuous flow, visual WIP limits, flexible iterations.	*Secondary* — Useful for operational maintenance post-launch.
Extreme Programming (XP)	Focuses heavily on engineering practices (TDD, pair programming).	*Partial* — Engineering practices adopted within Scrum Sprints.
Scaled (SAFe / LeSS)	Designed for large enterprise scaling across multiple teams.	*Overhead* — Unnecessary complexity for a single team.

Framework Recommendation

Scrum is recommended as the core methodology due to its structured sprint cadence, clear role accountability, and iterative milestone alignment for the Student Record Platform.